



Dimitris
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PhD

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English C2 | French B1 | Greek (native)
Permit C | Father of 2

Profile

Senior computational and experimental scientist with 10+ years of experience in translational research and an H-index of 9. Deep expertise in multi-omics data analysis, mechanistic toxicology, and molecular pathway elucidation. Proven track record leading integrative studies using transcriptomics, proteomics, and metabolomics to understand mechanisms of action.

Skills

Multi-Omics & Systems Biology

- High-throughput transcriptomics, proteomics, phosphoproteomics, metabolomics.
- Pathway and network analysis (IPA, Reactome, Cytoscape), including optimization-based modeling of signaling networks and logic reconstruction from perturbational omics data.
- Dataset integration and large-scale interpretation for mechanism discovery, including experience with high-dimensional perturbation-response data from platforms like LINCS.

Computational & Data Science

- R and Python for omics data analysis, statistical modeling, and biological network inference.
- Machine learning for biomarker discovery and drug target identification, unsupervised clustering of high-dimensional omics datasets.
- High-performance computing, Integer Linear Programming applied to modeling and solving biological network optimization problems.

Experimental Biology & Mechanistic Toxicology

- Experimental design and direction of non-clinical studies for mechanism-of-action elucidation.
- In vitro/ex vivo human and rodent models for toxicology, oxidative stress, and inflammation.
- Mammalian cell culture (2D, 3D), Luminex multiplex assays, RNA/DNA extraction, ELISA, xCelligence, several plate reader assays, compound spiking (including in 384-well plates).

Scientific Experience

May 2018 - Feb 2022 | Postdoctoral Fellow, **Philip Morris International**, Switzerland

- Led multi-omics studies on tobacco-related compounds to uncover molecular mechanisms. **Discovered several mechanisms of action** for anatabine and synthetically developed compounds with related pharmacological profiles, including a novel NRF2-activating effect.
- Applied pathway modeling for multi-omics interpretation and mechanism-of-action studies.
- Led projects that resulted in **patent filing**, internal business cases, and publications.

- Community: Recognized 5x with Above & Beyond the Call of Duty (ABCD) award including for pandemic response and for co-founding PMI's most successful mentoring program.

Jul 2016 - Nov 2016 | Data Scientist, **U.S. Food & Drug Administration (FDA)**, Washington DC

- Developed translational systems pharmacology models to predict drug-induced cardiotoxicity using transcriptomic data and integer linear programming.
- Improved predictive accuracy of existing models (from 79% to 88%) and published first-author results.
- Worked as an ORISE Fellow in FDA's Office of Clinical Pharmacology in a cross-disciplinary regulatory science setting.

Nov 2011 - Jan 2014 | Project Leader, **Protavio Ltd** (biotech startup), Athens, Greece

- Co-established systems pharmacology CRO; worked across R&D, operations, and client relations.
- Developed sample collection protocols used by major labs (Karolinska, Charite, University Hospital of Zürich).
- Delivered large-scale profiling projects to clients including Roche and Harvard.

Aug 2009 - Oct 2011 | Research Assistant, **Systems Bioengineering Lab NTUA**

- Contributed to systems biology studies of inflammation and drug response.
- Earned national undergraduate research award.

Recent Roles in Applied Science and Health AI

May 2025 – now | Scientific Business Development, **Protocol AI (Risklick AG)**, Switzerland

- Managing pharma accounts and supporting business development with scientific expertise.

Oct 2023 - May 2025 | Co-Founder & Scientific Lead, **Nuuro AG**, Switzerland

- Developed an AI-powered app for personalized nutrition.
- Built data models integrating user profiles with nutritional ontologies and health goals.
- Applied machine learning to enable real-time dietary decision support.

Mar 2022 - Sep 2023 | Founder & Scientific Lead, **Deepcare**, Switzerland

- Designed multi-modal checkups combining blood, microbiome, and epigenetic data.
- Built data interpretation pipelines to generate health insights from complex biomarker profiles.
- Gained experience turning complex biomedical data into user-friendly outcomes and engaging non-scientific stakeholders.

Education

Jan 2014 - Apr 2018 | **Ph.D. in Systems Biology**, National Technical University of Athens (NTUA)
Dissertation: Systems biology and pharmacology modeling of human disease mechanisms

Sep 2005 - Nov 2011 | **Diploma (MEng) in Mechanical Engineering**, NTUA, Greece